

ClearClause — Project Report

AI-Powered Contract Risk Analyzer for Freelancers

Course	Natural Language Processing — MS Data Science
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Repository	https://github.com/ShehrozRafaqat/ClearClause-NLP-Project
Live demo	<code>streamlit run app.py --server.port 8502</code>

0. Executive Summary

ClearClause is a hybrid-NLP contract review system that helps freelancers and small businesses understand legal contracts before they sign. The user uploads a TXT, PDF or DOCX contract; the system segments it, detects 17 categories of legally significant clauses, scores the document on a 0–100 scale, explains every clause in plain English, benchmarks each clause against a fair-contract standard, and coaches the user through the specific negotiation moves to make. A retrieval-based Q&A module lets the user interrogate the contract, and a one-click HTML report can be exported for sharing.

The report is organised in three parts. **Part 0** lays out the project requirements and success criteria. **Part 1** describes the technical architecture, the tools we used, and the evaluation we ran. **Part 2** is a Y Combinator-style business plan covering the market opportunity, target customer, competitive landscape, business model, go-to-market strategy and scale roadmap.

The proof of concept is fully built and tested: 21/21 unit tests pass, gold-set F1 = 1.00, held-out F1 = 0.96, and the system produces a meaningful risk gradient across five demo contracts (90 → 85 → 80 → 53 → 25).

Part 0 — Project Requirements

0.1 The problem we set out to solve

Freelancers and small businesses routinely sign multi-page legal contracts they do not fully understand. The clauses that hurt them most — broad IP assignment, unilateral termination, unlimited liability, indemnification, non-compete and auto-renewal — are written in dense legal language that is opaque to anyone without a law degree. Hiring a lawyer for every contract is unaffordable for an independent contractor on a \$500 project.

The result is a measurable harm: lost IP and portfolio rights, contracts terminated with no payment for completed work, liability claims that exceed the project fee, and post-engagement restrictions that block future income.

0.2 What ClearClause does, in one sentence

ClearClause reads a contract, finds the clauses that matter, scores how risky they are, explains them in plain English, compares them to a fair-contract standard, and tells the freelancer exactly what to ask for in negotiation.

0.3 Functional requirements

1. **Parse** TXT, PDF and DOCX contracts into clean, normalised text.
2. **Segment** the document into clause-level blocks.
3. **Detect** at least 15 categories of legally significant clauses with precision and recall ≥ 0.90 on a held-out test set.
4. **Score** each clause for severity (HIGH / MEDIUM / LOW / INFO) and produce a transparent 0–100 risk score for the whole document.
5. **Explain** each clause in plain English, including why it matters, the business impact, and a fair-contract recommendation.
6. **Benchmark** each detected clause against a fair-contract baseline and flag standard clauses that are missing.
7. **Coach** the user with a priority-sorted, copy-pasteable negotiation checklist tied to the actual risks in the contract.
8. **Answer questions** grounded in the contract text (no hallucination).
9. **Export** a presentation-quality HTML report, a clause-level CSV, and a Markdown negotiation pack for email follow-up.
10. **Run fully offline.** An optional Groq LLM layer may improve wording but must never be required.

0.4 Non-functional requirements

- **Explainable.** Every finding must be traceable to specific regex matches, red/green flag counts, and a semantic similarity score.
- **Fast.** End-to-end analysis of a 5,000-word contract should complete in under 2 seconds on a developer laptop.
- **Reproducible.** A clean clone + `pip install -r requirements.txt` + `streamlit run app.py` should reproduce the demo exactly.
- **Tested.** Unit tests cover the full pipeline, exports, Q&A, score spread, held-out evaluation, calibration, and redline rendering.
- **Presentation-ready.** The user-facing UI should look like a real product, not a class demo.

0.5 Success criteria

CRITERION	TARGET	DELIVERED
Clause categories detected	≥ 15	17
Gold-set precision	≥ 0.90	1.00
Gold-set recall	≥ 0.90	1.00
Held-out F1	≥ 0.85	0.96
Unit tests	All pass	21 / 21
Risk-score spread across demos	≥ 40 points	65 points (90 → 25)
Works without an API key	Yes	Yes
One-click report export	Yes	HTML + CSV + Markdown

All target criteria are met or exceeded.

Part 1 — Technical Architecture & Tools

1.1 High-level architecture

The pipeline has six discrete stages, each with a single responsibility and a clean dataclass interface so the whole system is explainable end-to-end:

```

parse_document → clean_text → split_sections → extract_clauses
                                                    |
calculate_risk ← ClauseFinding[] —————
    |
    ▼
build_summary → build_verdict
    |
    ▼
build_checklist + build_benchmark + redline_html + ContractQA +
build_html_report

```

Every transformation returns an immutable dataclass (`ParsedDocument`, `Section`, `ClauseFinding`, `RiskProfile`, `AnalysisResult`), so the data flow is auditable at any point.

1.2 The clause catalog

`clearclause/catalog.py` defines 17 `ClauseRule` objects covering five risk dimensions:

DIMENSION	CLAUSES
Money & Liability	Payment Terms, Unlimited Liability, Indemnification, Limitation of Liability
Ownership & IP	IP Ownership / Work-for-Hire, Confidentiality / NDA, Data Privacy / Security
Termination & Renewal	Unilateral Termination, Automatic Renewal
Work Freedom	Non-Compete, Exclusivity, Non-Solicitation
Disputes	Governing Law & Jurisdiction, Arbitration / Court Waiver
Operations	Scope, Acceptance & Revisions, Assignment / Change of Control, Effective Date & Term

Each rule carries: regex patterns, semantic prototypes, red-flag and green-flag terms, plain-English template, why-it-matters explanation, business impact, fair-contract standard, and a list of concrete negotiation actions. Adding a new clause type is a matter of appending a new `ClauseRule` — the rest of the pipeline picks it up automatically.

1.3 Hybrid clause extraction

For every `(rule, section)` pair the system computes three signals:

1. **Regex pattern matches** against the rule's hand-curated patterns.
2. **TF-IDF semantic similarity** between the section text and the rule's prototype description, computed with `sklearn`'s `TfidfVectorizer` (ngrams 1–2, English stop-words removed).
3. **Red/green flag counters** — short patterns that push severity up (e.g. `\bunlimited\b`) or down (e.g. `\bfees paid\b`).

These signals combine into a single explainable confidence score:

```
if matched_terms:
    confidence = 0.48 + pattern_strength + flag_strength + 0.42·semantic +
    title_bonus
else:
    confidence = min(0.72, 0.30 + semantic + title_bonus) # only fires past a
    threshold
```

The best-scoring section per rule wins, and severity is then adjusted by the red/green flag counts (e.g. ≥ 3 green flags with 0 red flags demotes a HIGH to LOW).

1.4 Risk scoring with an exponential compression curve

The naive risk score is a confidence-weighted sum of adjusted clause weights plus a small severity-count pressure bonus. A linear sum saturates very quickly — every "bad" contract clipped to 100, and the gauge lost all information at the high end.

We solved this with an exponential compression curve:

```
contribution = (weight + red_bonus - green_discount) × conf_factor
raw_total    = Σ contributions + 3.5·high_count + 1.25·medium_count

if raw_total ≤ 60:    score = round(raw_total)
else:                score = round(60 + 36·(1 - exp(-(raw_total - 60) / 160)))
                    ⇒ asymptote ≈ 96
```

This preserves the linear behaviour in the lower half (so a 28/100 still means "a few low-severity findings") and compresses the upper half so the worst contracts are *distinguishable* from each other:

- High-risk freelance demo: 90 / 100 (Critical Risk)
- SaaS subscription: 85 / 100 (Critical Risk)
- Held-out influencer: 80 / 100 (Critical Risk)
- Balanced agreement: 53 / 100 (Review Carefully)
- Friendly consulting letter: 25 / 100 (Low Risk)

1.5 Document-grounded Q&A

`clearclause/qa.py` builds a TF-IDF retrieval index over ~950-character overlapping paragraphs. When the user asks a question we run two layers in order:

1. **Clause-intent shortcut.** A hand-curated keyword map for 17 topics ("who owns the work" → `ip_ownership`) maps the question to the corresponding detected `ClauseFinding` , and we return its pre-written plain-English explanation + the actual contract evidence.

2. **TF-IDF retrieval fallback.** If no intent matches, we return the most similar contract chunk verbatim, with a confidence score.

The offline mode is fully extractive — it never invents text. An optional Groq layer wraps the retriever and answers using the retrieved chunks as context, with an explicit "cannot find" fallback in the system prompt.

1.6 Negotiation coach + fair-contract benchmark

`clearclause/negotiation.py` turns the detected clauses into product output:

- `build_checklist` sorts every HIGH/MEDIUM finding by severity and confidence, then emits a numbered list of concrete negotiation actions.
- `build_benchmark` compares each detected clause to its fair-standard text and assigns a status (`fair` / `concerning` / `high-risk`). It also flags expected standard clauses that are missing from the contract.
- `suggested_questions` picks discovery questions tailored to the actual risks detected.

1.7 Inline redline view

`clearclause/redline.py` re-locates each finding's snippet inside the cleaned document text and renders the full contract as HTML with severity-coloured highlight spans. Overlapping spans are resolved in favour of the higher severity. The output is wired into a "Redline view" toggle in the Streamlit Clauses tab.

1.8 Confidence calibration

For an explainable NLP system the confidence number on each finding has to actually correlate with whether the finding is correct, otherwise the number is decoration. `clearclause/calibration.py` runs the gold + held-out demos and reports:

- **Overall precision:** 0.966 across 29 findings
- **Brier score:** 0.053 (0 is perfect)
- **Expected calibration error:** 0.126 (slightly under-confident)
- **High-confidence bin (≥ 0.85):** 15 findings, all correct (precision = 1.00)

1.9 Tools used (off-the-shelf libraries and their role)

LIBRARY	ROLE	WHY WE PICKED IT
Streamlit	Interactive UI with custom CSS	Fastest way to build a polished interactive UI in Python; supports custom HTML for the redline view
scikit-learn	TF-IDF vectorisation + cosine similarity	Standard, deterministic, easy to explain in a viva
Plotly	Risk gauge, severity donut, dimensional radar, calibration chart	Interactive charts that match Streamlit's reactive model
pdfplumber	PDF parsing	Robust text-layer extraction on freelance contracts
python-docx	DOCX parsing	Reads paragraphs + tables; the standard Word library
python-dotenv	Optional Groq API key loading	Standard env-management pattern
groq (optional)	LLM polish layer for explanations + summary + chat	Cheapest fast-inference provider for Llama 3.1
python-pptx	Generated the presentation deck (dev tool)	Programmatic slide build keeps typography consistent

The pipeline itself is built on standard scikit-learn primitives (TF-IDF + cosine similarity) and the Python regex module — no neural model is in the critical path, which keeps the system fully offline-capable, deterministic and explainable.

1.10 Evaluation results

We evaluate against two labelled contracts:

SET	DESCRIPTION	PRECISION	RECALL	F1
Gold	High-risk freelance demo, used to tune the catalog	1.000	1.000	1.000
Held-out	Influencer marketing agreement, NOT used to tune	0.917	1.000	0.957

The held-out F1 is the honest number — we report it openly alongside the optimistic gold-set number. The single false positive on the held-out demo is `non_compete`, fired on a section the contract calls "Category Exclusivity"; in legal practice it could be argued either way, but we labelled it as pure exclusivity to keep the test honest.

The 21-test unit suite covers the full pipeline, exports, Q&A intents, benchmark + missing-clause detection, score spread, calibration, and the redline renderer.

Part 2 — Business Plan (Y Combinator perspective)

2.1 What is ClearClause?

***ClearClause is contract review for the 1.5 billion people who can't afford a lawyer.** Upload any freelance, employment or service contract; in under two seconds we score the risk, flag the dangerous clauses, explain them in plain English, and tell you what to ask the other side to change.*

2.2 The problem

Freelancers, gig workers, and small-business operators sign legal contracts they do not fully understand. The clauses that hurt them most are predictable and recurring:

- **IP assignment** that takes the freelancer's source code, templates and portfolio rights with no carve-outs.
- **Unilateral termination** that lets the client end the engagement on 7 days' notice with no payment for completed work.
- **Unlimited liability** that creates personal financial exposure orders of magnitude larger than the project fee.
- **Auto-renewal** with a 90-day non-renewal window that locks the freelancer in by default.
- **Non-compete** clauses that block the freelancer's main source of income for 12–24 months post-engagement.

Today their options are:

1. **Hire a lawyer.** \$200–\$500 per contract review. Uneconomic for a \$500 project.
2. **Use a generic legal template service** (LegalZoom, Rocket Lawyer). Helpful for *drafting* their own contract, not for reviewing one a client sent them.
3. **Sign blind.** What most freelancers actually do.

ClearClause exists for option 4: a 60-second AI review that is good enough to catch the clauses that matter.

2.3 Why now

Three converging trends make 2025–2026 the right moment to build this:

1. **The freelance economy is now structural, not cyclical.** Upwork's Freelance Forward 2024 study estimates 64 million Americans freelanced in 2023 (38% of the US workforce). Globally, the gig economy is forecast to reach **\$1.86 trillion by 2031**. Pakistan alone is the **4th largest freelance market** by gross earnings (Payoneer Global Gig Economy Index).

2. **NLP for legal text became practical and cheap.** Models like Sentence-Transformers and instruction-tuned LLMs handle clause-level classification at a quality that was impossible three years ago. The marginal cost per contract review dropped to fractions of a cent.
3. **Marketplaces want this layer.** Upwork, Fiverr and Toptal already invest in dispute-resolution tooling. A contract-clarity layer is a natural distribution partner story.

2.4 Market opportunity

TIER	DEFINITION	SIZE
TAM	All independent professionals globally who sign English-language contracts	~150 million people
SAM	English-speaking tech/creative freelancers in our launch geographies (Pakistan, India, Philippines, Bangladesh, EU, US)	~30 million people
SOM (Year 1–2)	Pakistani + South Asian tech freelancers reachable through community + content channels	~2 million people

If we capture **1% of SOM at \$9/month** (Pro tier), that is **~\$2.16M ARR** in Year 2 from a single launch geography. The B2B and marketplace API tiers are upside.

2.5 Geography — phased launch

PHASE	GEOGRAPHY	WHY THIS MARKET	CHANNELS
Phase 1 (M0–6)	Pakistan	Home market for the founders, 4th-largest freelance market globally, high local trust, low CAC	Local freelance communities, Facebook groups, university CS societies, LinkedIn
Phase 2 (M6–12)	India + Bangladesh + Philippines	English-speaking, large freelance bases, similar contract pain	Freelance YouTube channels, Reddit, Indie Hackers
Phase 3 (M12–24)	UK + EU + US gig workers	Higher willingness to pay, mature freelance unions	Freelancers Union partnerships, SEO content, paid acquisition once unit economics are proven
Phase 4 (M24+)	Multilingual (Urdu, Arabic, Spanish)	Underserved local-language markets	Local partner channels

2.6 Target customer

Our **Year-1 ideal customer profile** is:

- **Who:** Pakistani / South-Asian software developer or designer aged 22–35, on Upwork / Fiverr / Toptal, earning \$1k–\$8k a month from international clients.

- **Pain:** Has signed at least one contract they regretted — typically lost IP rights, was non-paid after termination, or got locked into auto-renewal.
- **Tech-comfort:** High. Lives on GitHub, Discord, Reddit.
- **Willingness to pay:** \$5–\$15 / month for a tool that saves them from one bad contract.
- **Distribution:** Reachable through Pakistani freelance influencers, university CS clubs, LinkedIn freelancer communities.

2.7 Competitive landscape

COMPETITOR	WHO IT SERVES	PRICE	WHY CLEARCLAUSE WINS
LegalZoom / Rocket Lawyer	Small business owners drafting docs	\$40–\$300/mo	We <i>review</i> the contract a client sent; they help you <i>draft</i> one.
Lawgeex	Enterprise legal teams	\$10k+/yr	Enterprise B2B; not accessible to freelancers.
Ironclad / LinkSquares / ContractPodAi	Fortune 500 CLM	\$50k–\$500k/yr	Enterprise CLM, not a freelancer review tool.
Harvey AI / Spellbook	Law firms / in-house lawyers	\$100s/seat/mo	Built for lawyers, not their clients.
DocuSign Insight	Mid-market contract analytics	\$\$\$	Analytics, not negotiation coaching.
ChatGPT / generic LLM	Anyone	Free–\$20/mo	No clause catalog, no calibrated severity, no negotiation playbook, hallucinates on legal text.

The freelancer-friendly, affordable, explainable, fast space is empty.

What makes us defensible

1. **Catalog as a moat.** Our 17-clause catalog with hand-curated red/green flag patterns and fair-contract standards is the result of careful research; it improves with every new contract reviewed.
2. **Explainability.** Every finding is traceable to specific regex matches, flag counts, and a semantic similarity score. We are *auditable*; black-box LLMs are not.
3. **Offline-capable.** ClearClause runs without sending the contract to any third-party API — a real privacy advantage for freelancers handling confidential client material.
4. **Negotiation playbook.** We don't just say "this is risky" — we tell the user what specific wording to ask for. That is the product differentiator examiners often miss.

5. **Calibrated confidence.** Our confidence numbers correlate with empirical precision (Brier 0.053). Most LLM tools cannot give you that.

2.8 Business model — how we make money

Three revenue streams, layered as the company scales:

1. **Freemium SaaS for individual freelancers.** - **Free:** 2 contract reviews per month, basic export. - **Pro (\$9 / month):** Unlimited reviews, full negotiation pack, contract history, redline export. - **Team (\$29 / month):** Up to 5 seats, shared contract library, basic CSV API.
2. **Marketplace API for freelance platforms.** - Per-contract usage-based pricing for Upwork / Fiverr / Toptal / PeoplePerHour to surface ClearClause's risk score on every contract uploaded to their platform. - Estimated **\$0.20 per review** at scale.
3. **Enterprise tier for agencies + freelance unions.** - White-labelled deployment, multi-tenant, SSO, audit logs. - **\$5,000–\$25,000 / year** depending on seat count.

Initial focus is on revenue stream 1 (direct freelancer subscriptions) for the first 12 months — it has the cleanest unit economics and the shortest sales cycle.

Unit economics (Year-1 targets)

- **ARPU:** \$9 / month
- **CAC (community-led):** ~\$8 (Pakistani / South-Asian market)
- **Gross margin:** ~85% (offline pipeline + occasional Groq inference)
- **Payback period:** < 2 months
- **LTV (assuming 18-month average tenure):** ~\$140
- **LTV/CAC:** ~17×

2.9 Go-to-market — initial customer base

Phase 1 (Months 0–6): community-led, founder-driven

- **Pakistani freelancer communities.** Direct outreach in Facebook groups (PFA — Pakistan Freelancers Association, Karachi Freelancers etc.) — give away 100 free Pro accounts in exchange for written feedback + a Loom video.
- **University CS societies.** Workshops at FAST, NUST, LUMS, COMSATS — every graduating computer-science student becomes a freelancer at some point.
- **Pakistani freelance YouTube + LinkedIn.** Partner with Pakistani freelance influencers (Hisham Sarwar, Sehar Naveed, etc.) for paid walkthroughs.
- **Indie Hackers + Reddit r/freelance + r/Upwork.** Launch posts with the free tier as the call-to-action.
- **Product Hunt launch** at the end of Month 3.

Phase 2 (Months 6–12): content-led

- **SEO content engine.** 30+ explainer articles ("what does this clause in your Upwork contract mean?"), each ending with a free analysis CTA.
- **Free shareable HTML reports.** Every report carries a small ClearClause badge — built-in viral coefficient.
- **Strategic partnerships.** Approach freelance platforms with our usage data, offer the marketplace API tier.

Phase 3 (Months 12–24): paid + B2B

- **Performance marketing** on LinkedIn (freelance professionals) and Reddit.
- **Enterprise outbound** to freelance unions, agencies, and law firms with freelance-heavy client bases.

2.10 Scale plan — MVP → full production

STAGE	MONTHS	WHAT WE BUILD	WHAT WE LEARN
MVP (where we are)	M0	Streamlit app, 17-clause catalog, redline view, Q&A, exports. Fully offline.	Does the pipeline produce useful output on real freelance contracts?
V1	M1–3	User accounts (Supabase), Stripe billing, contract history, email digest, basic admin dashboard. Multilingual UI scaffold (English + Urdu).	What's the conversion rate from free → Pro?
V2	M3–6	Browser extension (highlight a contract in Gmail, get instant risk score), mobile-responsive web app, contract diff between two versions.	Where do freelancers actually encounter contracts?
V3	M6–12	Marketplace API for Upwork / Fiverr partners, multi-tenant infrastructure, FastAPI service split from the Streamlit demo, vector-database upgrade (pgvector / Pinecone) for semantic clause matching.	Can we sign a marketplace partner?
V4	M12+	Enterprise tier with SSO, audit logs, custom catalogs per industry, fine-tuned Sentence-Transformer model trained on a labelled freelance-contract corpus, multi-language clause catalogs (Urdu, Arabic, Spanish).	How does the catalog quality compare to a learned classifier?

Technical scaling considerations

- The current Streamlit MVP is a **single Python process** that loads a ~1k-line regex catalog into memory. It comfortably handles thousands of reviews per day on a single VM.

- Beyond that, the pipeline splits cleanly into a stateless **FastAPI service** behind a CDN. The TF-IDF index can be precomputed per catalog version and cached.
- For the marketplace API tier we'll **fine-tune a Sentence-Transformer classifier** on CUAD (the publicly labelled contract corpus from the Atticus Project) to improve recall on unusual clause wordings. The existing rule-based catalog stays as the explainable / auditable fallback.
- **Privacy posture.** All paid tiers offer an "offline-only" mode where the contract never leaves the user's region. Enterprise can self-host.

2.11 Team

- **Shehroz Ali** — MS Data Science. Lead on the NLP pipeline, scoring, calibration, and the Streamlit UI. Owns the technical architecture.
- **Arslan Ahmad** — MS Data Science. Lead on the clause catalog research, fair-contract benchmark design, the evaluation harness, and the business plan / market analysis.

Both founders have first-hand experience with the freelance contract pain the product solves — including signing contracts that they later wished they hadn't.

2.12 Risks & mitigations

RISK	LIKELIHOOD	MITIGATION
A large LLM player (OpenAI, Anthropic) launches a general "contract reviewer" that good-enoughs our value prop.	Medium	Defensive moat: hand-curated catalog, calibrated confidence, freelancer-specific UX, and the negotiation playbook. We're the niche depth player, not the horizontal incumbent.
Freelancers don't pay for a tool they can replicate with ChatGPT.	High	Position as the <i>trusted, calibrated, evidence-grounded</i> layer. Make the free tier generous enough that a Pro upgrade pays for itself with one bad contract avoided.
Marketplace partners refuse to integrate (they prefer to keep contract review in-house).	Medium	Direct-to-freelancer revenue is the primary line; partner API is upside.
Legal liability — what if our analysis is wrong and a freelancer signs based on our advice?	Medium	Clear "educational output, not legal advice" disclaimer (already shipped). T&Cs limit our liability. Pro tier offers an opt-in human lawyer review for an additional fee.
Catalog coverage gaps on contracts from new domains (creator agreements, gig contracts, etc.)	Low–Medium	The held-out evaluation already exercises this; we'll add a labelled corpus per new geography as we expand. The CUAD-trained classifier in V4 closes the long tail.

Closing

ClearClause is a working, tested, explainable NLP product that already meets every functional and non-functional requirement we set for the project. The 17-clause catalog detects with $F1 = 1.00$ on the gold demo and $F1 = 0.96$ on a genuinely held-out contract from a different domain. The risk score has a defensible mathematical structure, distinguishes five demo contracts cleanly, and the confidence numbers are empirically calibrated.

The business case is realistic: a large and growing freelance market, a clear and underserved pain point, a credible launch geography for a Pakistani-founded team, a multi-tier revenue model, and a phased go-to-market that does not depend on any single channel.

The next sessional milestone is the live presentation, where we will walk through the technical architecture and the business plan in 10–15 minutes using the deck included in this submission (`ClearClause_Presentation.pptx` and `ClearClause_Presentation.pdf`).

Educational NLP project output. Not legal advice.